COLECCIÓN CANÓNICA

GRAMÁTICA

$E' \longrightarrow E\$$
$E \longrightarrow E+T$
$E \longrightarrow T$
$T \longrightarrow T*F$
$T \longrightarrow F$
$F \longrightarrow (E)$

cerradura $(\{[E' \longrightarrow \cdot E\$]\})$

$I_0 = \{[E' \longrightarrow \cdot E\$][E \longrightarrow \cdot E+T][E \longrightarrow \cdot T][T \longrightarrow \cdot T*F]$
$[T \longrightarrow \cdot F][F \longrightarrow \cdot(E)][F \longrightarrow \cdot id]\}$

$Ir_a(I_0, E) =$ cerradura $(\{[E' \longrightarrow E \cdot \$][E \longrightarrow E \cdot +T]\})$

$I_1 = \{[E' \longrightarrow E \cdot \$][E \longrightarrow E \cdot +T]\}$

$Ir_a(I_0, T) =$ cerradura $(\{[E \longrightarrow T \cdot][T \longrightarrow T \cdot *F]\})$

$I_2 = \{[E \longrightarrow T \cdot][T \longrightarrow T \cdot *F]\})$

$Ir_a(I_0, F) =$ cerradura $(\{[T \longrightarrow F \cdot]\})$

$I_3 = \{[T \longrightarrow F \cdot]\}$

$Ir_a(I_0, ( ) =$ cerradura $(\{[F \longrightarrow (\cdot E)]\})$

$I_4 = \{[F \longrightarrow (\cdot E)][E \longrightarrow \cdot E+T][E \longrightarrow \cdot T][T \longrightarrow \cdot T*F]$
$[T \longrightarrow \cdot F][F \longrightarrow \cdot(E)][F \longrightarrow \cdot id]\}$

$Ir_a(I_0, id) =$ cerradura $(\{[F \longrightarrow id \cdot]\})$

$I_5 = \{[F \longrightarrow id \cdot]\}$

$Ir_a(I_1, \$) =$ Aceptación

$Ir_a(I_1, +) =$ cerradura $(\{[E \longrightarrow E+\cdot T]\})$

$I_6 = \{[E \longrightarrow E+\cdot T][T \longrightarrow \cdot T*E][T \longrightarrow \cdot F][F \longrightarrow \cdot(E)][F \longrightarrow \cdot id]\}$

$Ir_a(I_2, *) =$ cerradura $(\{[T \longrightarrow T* \cdot F]\})$

$I_7 = \{[T \longrightarrow T* \cdot F][F \longrightarrow \cdot(E)][F \longrightarrow \cdot id]\}$

$Ir_a(I_4, E) =$ cerradura $(\{[F \longrightarrow (E \cdot)][E \longrightarrow E \cdot +T]\})$

$I_8 = \{[F \longrightarrow (E \cdot)][E \longrightarrow E \cdot +T]\}$

$Ir_a(I_4, T) =$ cerradura $(\{[E \longrightarrow T \cdot][T \longrightarrow F \cdot *F]\}) = I_2$

$Ir_a(I_4, F) =$ cerradura $(\{[T \longrightarrow F \cdot]\}) = I_3$

$Ir_a(I_4, ()$ = cerradura $(\{[F \rightarrow (\cdot E)]\})$ $= I_4$

$Ir_a(I_4, id)$ = cerradura $(\{[F \rightarrow id \cdot]\})$ $= I_5$

$Ir_a(I_6, T)$ = cerradura $(\{[E \rightarrow E+T \cdot][T \rightarrow T \cdot * F]\})$

$I_9 = \{[E \rightarrow E+T \cdot][T \rightarrow T \cdot * F]\}$

$Ir_a(I_6, F)$ = cerradura $(\{[T \rightarrow F \cdot]\})$ $= I_3$

$Ir_a(I_6, ()$ = cerradura $(\{[F \rightarrow (\cdot E)]\})$ $= I_4$

$Ir_a(I_6, id)$ = cerradura $(\{[F \rightarrow id \cdot]\})$ $= I_5$

$Ir_a(I_7, F)$ = cerradura $(\{[T \rightarrow T * F \cdot]\})$

$I_{10} = \{[T \rightarrow T * F \cdot]\}$

$Ir_a(I_7, ()$ = cerradura $(\{[F \rightarrow (\cdot E)]\})$ $= I_4$

$Ir_a(I_7, id)$ = cerradura $(\{[F \rightarrow id \cdot]\})$ $= I_5$

$Ir_a(I_8,))$ = cerradura $(\{[F \rightarrow (E) \cdot]\})$

$I_{11} = \{[F \rightarrow (E) \cdot]\}$

$Ir_a(I_8, +)$ = cerradura $(\{[E \rightarrow E + \cdot T]\}) = I_6$

$Ir_a(I_9, *)$ = cerradura $(\{[T \rightarrow T * \cdot F]\}) = I_7$

GRAMATICA 2.

D' → D
D → DD
D → T id L ;
D → , id L
L → ε
T → float
T → int

cerradura({[D' → • D $]})

$I_0 = \{[D' \to \bullet D\$][D \to \bullet DD][D \to \bullet T\ id\ L;]$
$[T \to \bullet float][T \to \bullet int]\}$

$I_{r_a}(I_0, D) = cerradura(\{[D' \to D \bullet \$][D \to D \bullet D]\})$

$I_1 = \{[D' \to D \bullet \$][D \to D \bullet D][D \to \bullet DD]$
$[D \to \bullet T\ id\ L;][T \to float][T \to \bullet int]\}$

$I_{r_a}(I_0, T) = cerradura(\{[D \to T \bullet id\ L;]\})$

$I_2 = \{[D \to T \bullet id\ L;]\}$

$I_{r_a}(I_0, float) = cerradura(\{[T \to float \bullet]\})$

$I_3 = \{[T \to float \bullet]\}$

$I_{r_a}(I_0, int) = cerradura(\{[T \to int \bullet]\})$

$I_4 = \{[T \to int \bullet]\}$

$I_{r_a}(I_1, \$) = Aceptación$

$I_{r_a}(I_1, D) = cerradura(\{[D \to DD \bullet][D \to D \bullet D]\})$

$I_5 = \{[D \to DD \bullet][D \to D \bullet D][D \to \bullet DD][D \to \bullet T\ id\ L;]$
$[T \to \bullet float][T \to \bullet int]\}$

$I_{r_a}(I_1, T) = cerradura(\{[D \to T \bullet id\ L;]\}) = I_2$

$I_{r_a}(I_1, float) = cerradura(\{[T \to float \bullet]\}) = I_3$

$I_{r_a}(I_1, int) = cerradura(\{[T \to int \bullet]\}) = I_4$

$I_{r_a}(I_2, id) = cerradura(\{[D \to T\ id \bullet L;]\})$

$I_6 = \{[D \to T\ id \bullet L;][L \to \bullet, id\ L][L \to \bullet]\}$

$I_{r_a}(I_5, D) = cerradura(\{[D \to DD \bullet][D \to D \bullet D]\}) = I_5$

$I_{r_a}(I_5, T) = cerradura(\{[D \to T \bullet id\ L;]\}) = I_2$

$Ir_a(I_5, float) = cerradura(\{[T \rightarrow float \cdot]\}) = I_3$

$Ir_a(I_5, int) = cerradura(\{[T \rightarrow int \cdot]\}) = I_4$

$Ir_a(I_6, L) = cerradura(\{[D \rightarrow T\ id\ L \cdot ;]\})$

$I_7 = \{[D \rightarrow T\ id\ L \cdot ;]\}$

$Ir_a(I_6, ,) = cerradura(\{[L \rightarrow , \cdot id\ L]\})$

$I_8 = \{[L \rightarrow , \cdot id\ L]\}$

$Ir_a(I_7, ;) = cerradura(\{[D \rightarrow T\ id\ L ; \cdot]\})$

$I_9 = \{[D \rightarrow T\ id\ L ; \cdot]\}$

$Ir_a(I_8, id) = cerradura(\{[L \rightarrow , id \cdot L]\})$

$I_{10} = \{[L \rightarrow , id \cdot L][L \rightarrow \cdot , id\ L][L \rightarrow \cdot]\}$

$Ir_a(I_{10}, L) = cerradura(\{[L \rightarrow , id\ L \cdot]\})$

$I_{11} = \{[L \rightarrow , id\ L \cdot]\}$

$Ir_a(I_{10}, ,) = cerradura(\{[L \rightarrow , \cdot id\ L]\}) = I_8$

GRAMATICA 3

D' → D$
D → T L;
L → id L
L → , id L
L → ε
T → float
T → int
T → ε

cerradura({[D' → · D$]})

$I_0 = \{[D' \to \cdot D\$][D \to \cdot TL;][T \to \cdot float][T \to \cdot int][T \to \cdot]\}$

$I_1 = \{[D' \to D \cdot \$]\}$

$Ir_a(I_0, T) = cerradura(\{[D \to T \cdot L;]\})$

$I_2 = \{[D \to T \cdot L;][L \to \cdot id\ L][L \to \cdot, id\ L][L \to \cdot]\}$

$Ir_a(I_0, float) = cerradura(\{[T \to float \cdot]\})$

$I_3 = \{[T \to float \cdot]\}$

$Ir_a(I_0, int) = cerradura(\{T \to int \cdot]\})$

$I_4 = \{[T \to int \cdot]\}$

$Ir_a(I_1, \$) = Aceptación$

$Ir_a(I_2, L) = cerradura(\{[D \to T\ L \cdot ;]\})$

$I_5 = \{[D \to T\ L \cdot ;]\}$

$Ir_a(I_2, id) = cerradura(\{[L \to id \cdot L]\})$

$I_6 = \{[L \to id \cdot L][L \to \cdot, id\ L][L \to \cdot]\}$

$Ir_a(I_2, ,) = cerradura(\{[L \to , \cdot id\ L]\})$

$I_7 = \{[L \to , \cdot id\ L]\}$

$Ir_a(I_5, ;) = cerradura(\{[D \to T\ L ; \cdot]\})$

$I_8 = \{[D \to T\ L ; \cdot]\}$

$Ir_a(I_6, L) = cerradura(\{[L \to id\ L \cdot]\})$

$I_9 = \{[L \to id\ L \cdot]\}$

$Ir_a(I_6, id) = cerradura(\{[L \to id \cdot L]\}) = I_6$

$I_{r-a}(I_6, ,) = \text{cerradura}(\{[L \rightarrow , \cdot id\ L]\}) = I_7$

$I_{r-a}(I_7, id) = \text{cerradura}(\{[L \rightarrow , id \cdot L]\})$

$I_{10} = \{[L \rightarrow , id \cdot L][L \rightarrow \cdot id\ L][L \rightarrow \cdot , id L][L \rightarrow \cdot]\}$

$I_{r-a}(I_{10}, L) = \text{cerradura}(\{[L \rightarrow , id\ L \cdot]\})$

$I_{11} = \{[L \rightarrow , id\ L \cdot]\}$

$I_{r-a}(I_{10}, id) = \text{cerradura}(\{[L \rightarrow id \cdot L]\}) = I6$

$I_{r-a}(I_{10}, ,) = \text{cerradura}(\{[L \rightarrow , \cdot id\ L]\}) = I_7$

GRAMATICA 4

$S' \rightarrow S\ \$$
$S \rightarrow S\ ;\ S$
$S \rightarrow id := E$
$S \rightarrow print\ (L)$
$E \rightarrow id$
$E \rightarrow num$
$E \rightarrow E + E$
$E \rightarrow (S, E)$
$L \rightarrow E$
$L \rightarrow L, E$

cerradura $(\{[S' \rightarrow \cdot S\$]\})$

$I_0 = \{[S' \rightarrow \cdot S\$][S \rightarrow \cdot S;S][S \rightarrow \cdot id := E]$
$[S \rightarrow \cdot print\ (L)]\}$

$I_{r_a}(I_0, S) =$ cerradura $(\{[S' \rightarrow S \cdot \$][S \rightarrow S \cdot ;S]\})$

$I_1 = \{[S' \rightarrow S \cdot \$][S \rightarrow S \cdot ;S]\}$

$I_{r_a}(I_0, id) =$ cerradura $(\{[S \rightarrow id \cdot := E]\})$

$I_2 = \{[S \rightarrow id \cdot := E]\}$

$I_{r_a}(I_0, print) =$ cerradura $(\{[S \rightarrow print \cdot (L)]\})$

$I_3 = \{[S \rightarrow print \cdot (L)]\}$

$I_{r_a}(I_1, \$) =$ Aceptación

$I_{r_a}(I_1, ;) =$ cerradura $(\{[S \rightarrow S\ ; \cdot S]\})$

$I_4 = \{[S \rightarrow S\ ; \cdot S][S \rightarrow \cdot S;S][S \rightarrow \cdot id := E][S \rightarrow \cdot print\ (L)]\}$

$I_{r_a}(I_2, :=) =$ cerradura $(\{[S \rightarrow id := \cdot E]\})$

$I_5 = \{[S \rightarrow id := \cdot E][E \rightarrow \cdot id][E \rightarrow \cdot num][E \rightarrow \cdot E + E][E \rightarrow \cdot (S,E)]\}$

$I_{r_a}(I_3, () =$ cerradura $(\{[S \rightarrow print\ (\cdot L)]\})$

$I_6 = \{[S \rightarrow print\ (\cdot L)][L \rightarrow \cdot E][L \rightarrow \cdot L, E][E \rightarrow \cdot id]$
$[E \rightarrow \cdot num][E \rightarrow E + E][E \rightarrow \cdot (S,E)]\}$

$I_{r_a}(I_4, S) =$ cerradura $(\{[S \rightarrow S\ ; S \cdot][S \rightarrow S \cdot ; S]\})$

$I_7 = \{[S \rightarrow S\ ; S \cdot][S \rightarrow S \cdot ; S]\}$

$I_{r_a}(I_4, id) =$ cerradura $(\{[S \rightarrow id \cdot := E]\}) = I_2$

$I_{r_a}(I_4, print) =$ cerradura $(\{[S \rightarrow print \cdot (L)]\}) = I_3$

$I_{r_a}(I_5, E) =$ cerradura $(\{[S \rightarrow id := E \cdot][E \rightarrow E \cdot + E]\})$

$I_8 = \{[S \rightarrow id := E \cdot][E \rightarrow E \cdot + E]\}$

Printaform

$I_{r_a}(I_5, id) = \text{cerradura}(\{[E \to id \cdot]\})$

$I_9 = \{[E \to id \cdot]\}$

$I_{r_a}(I_5, num) = \text{cerradura}(\{[E \to num \cdot]\})$

$I_{10} = \{[E \to num \cdot]\}$

$I_{r_a}(I_5, () = \text{cerradura}(\{[E \to (\cdot S, E)]\})$

$I_{11} = \{[E \to (\cdot S, E)][S \to \cdot S ; S][S \to \cdot id := E]$

$[S \to \cdot print(L)]\}$

$I_{r_a}(I_6, L) = \text{cerradura}(\{[S \to print(L \cdot)][L \to L \cdot, E]\})$

$I_{12} = \{[S \to print(L \cdot)][L \to L \cdot, E]\}$

$I_{r_a}(I_6, E) = \text{cerradura}(\{[L \to E \cdot][E \to E \cdot + E]\})$

$I_{13} = \{[L \to E \cdot][E \to E \cdot + E]\}$

$I_{r_a}(I_6, id) = \text{cerradura}(\{[E \to id \cdot]\}) = I_9$

$I_{r_a}(I_6, num) = \text{cerradura}(\{[E \to num \cdot]\}) = I_{10}$

$I_{r_a}(I_6, () = \text{cerradura}(\{[E \to (\cdot S, E)]\}) = I_{11}$

$I_{r_a}(I_7, ;) = \text{cerradura}(\{[E \to (\cdot S, E)]\}) = I_4$

$I_{r_a}(I_8, +) = \text{cerradura}(\{[E \to E + \cdot E]\})$

$I_{14} = \{[E \to E + \cdot E][E \to \cdot id][E \to \cdot num][E \to \cdot E + E]$

$[E \to \cdot (S, E)]\}$

$I_{r_a}(I_{11}, S) = \text{cerradura}(\{[E \to (S \cdot, E)][S \to S \cdot ; S]\})$

$I_{15} = \{[E \to (S \cdot, E)][S \to S \cdot ; S]\}$

$I_{r_a}(I_{11}, id) = \text{cerradura}(\{[S \to id \cdot := E]\}) = I_2$

$I_{r_a}(I_{11}, print) = \text{cerradora}(\{[S \to print \cdot (L)]\}) = I_3$

Printaform

$I_{r-a}(I_{12},)) = \text{cerradura}(\{[S \to \text{print}(L)\cdot]\})$

$I_{16} = \{[S \to \text{print}(L)\cdot]\}$

$I_{r-a}(I_{12},,) = \text{cerradura}(\{[L \to L,\cdot E]\})$

$I_{17} = \{[L \to L,\cdot E][E \to \cdot id][E \to \cdot num][E \to \cdot E+E]$

$[E \to \cdot(S,E)]\}$

$I_{r-a}(I_{13},+) = \text{cerradura}(\{[E \to E+\cdot E]\}) = I_{14}$

$I_{r-a}(I_{14},E) = \text{cerradura}(\{[E \to E+E\cdot][E \to E\cdot+E]\})$

$I_{18} = \{[E \to E+E\cdot][E \to E\cdot+E]\}$

$I_{r-a}(I_{14},id) = \text{cerradura}(\{[E \to id\cdot]\}) = I_9$

$I_{r-a}(I_{14},num) = \text{cerradura}(\{[E \to num\cdot]\}) = I_{10}$

$I_{r-a}(I_{14},() = \text{cerradura}(\{[E \to (\cdot S,E)]\}) = I_{11}$

$I_{r-a}(I_{15},,) = \text{cerradura}(\{[E \to (S,\cdot E)]\})$

$I_{19} = \{[E \to (S,\cdot E)][E \to \cdot id][E \to \cdot num]$

$[E \to \cdot E+E][E \to \cdot(S,E)]\}$

$I_{r-a}(I_{15},;) = \text{cerradura}(\{[S \to S;\cdot S]\}) = I_4$

$I_{r-a}(I_{17},E) = \text{cerradura}(\{[L \to L,E\cdot][E \to E\cdot+E]\})$

$I_{20} = \{[L \to L,E\cdot][E \to E\cdot+E]\}$

$I_{r-a}(I_{17},id) = \text{cerradura}(\{[E \to id\cdot]\}) = I_9$

$I_{r-a}(I_{17},num) = \text{cerradura}(\{[E \to num\cdot]\}) = I_{10}$

$I_{r-a}(I_{17},() = \text{cerradura}(\{[E \to (\cdot S,E)]\}) = I_{11}$

$I_{r-a}(I_{18},+) = \text{cerradura}(\{[E \to E+\cdot E]\}) = I_{14}$

$I_{r-a}(I_{19},E) = \text{cerradura}(\{[E \to (S,E\cdot)][E \to E\cdot+E]\})$

$I_{21} = \{[E \to (S,E\cdot)][E \to E\cdot+E]\}$

$I_{r-a}(I_{19},id) = \text{cerradura}(\{[E \to id\cdot]\}) = I_9$

$I_{r_a}(I_{20}, +) = \text{cerradura}(\{[E \rightarrow E + \cdot E]\}) = I_{14}$

$I_{r_a}(I_{21},)) = \text{cerradura}(\{[E \rightarrow (S, E) \cdot]\})$

$I_{22} = \{[E \rightarrow (S, E) \cdot]\}$

$I_{r_a}(I_{21}, +) = \text{cerradura}(\{[E \rightarrow E + \cdot E]\}) = I_{14}$

GRAMATICA 5

S' → S $
S → T V ;
V → id V'
V' → [nint] V'
V' → ε
T → int
T → char
T → Float

cerradura ({[S' → • S $]})

$I_0 = \{[S' \to \bullet S\$][S \to \bullet T V ;][T \to \bullet int][T \to \bullet char][T \to \bullet Float]\}$

$Ir_a(I_0, S) = cerradura(\{[S' \to S \bullet \$]\})$

$I_1 = \{[S' \to S \bullet \$]\}$

$Ir_a(I_0, T) = cerradura(\{[S \to T \bullet V ;]\})$

$I_2 = \{[S \to T \bullet V ;][V \to \bullet id\ V']\}$

$Ir_a(I_0, int) = cerradura(\{[T \to int \bullet]\})$

$I_4 = \{[T \to char \bullet]\}$

$Ir_a(I_0, Float) = cerradura(\{[T \to Float \bullet]\})$

$I_5 = \{[T \to Float \bullet]\}$

$Ir_a(I_1, \$) = Aceptación$

$Ir_a(I_2, V) = cerradura(\{[S \to T V \bullet ;]\})$

$I_6 = \{[S \to T V \bullet ;]\}$

$Ir_a(I_2, id) = cerradura(\{[V \to id \bullet V']\})$

$I_7 = \{[V \to id \bullet V'][V' \to \bullet [nint]\ V'][V' \to \bullet]\}$

$Ir_a(I_6, ;) = cerradura(\{[S \to T\ V\ ; \bullet]\})$

$I_8 = \{[S \to T\ V\ ; \bullet]\}$

$Ir_a(I_7, V') = cerradura(\{[V \to id\ V' \bullet]\})$

$I_9 = \{[V \to id\ V' \bullet]\}$

$Ir_a(I_7, [) = cerradura(\{[V' \to [\bullet nint]\ V']\})$

$I_{10} = \{[V' \to [\bullet nint]\ V']\}$

$Ir_a(I_{10}, nint) = cerradura(\{[V' \to [nint \bullet]\ V']\})$

$I_{11} = \{[V' \to [nint \bullet]\ V']\}$

$I_{r_a}(I_{11},]) = \text{cerradura}(\{[V' \rightarrow [\text{nint}] \cdot V']\})$

$I_{12} = \{[V' \rightarrow [\text{nint}] \cdot V'][V' \rightarrow \cdot [\text{nint}]\ V'][V' \rightarrow \cdot]\}$

$I_{r_a}(I_{12}, V') = \text{cerradura}(\{[V' \rightarrow [\text{nint}]\ V' \cdot]\})$

$I_{13} = \{[V' \rightarrow [\text{nint}]\ V' \cdot]\}$

$I_{r_a}(I_{12}, [) = \text{cerradura}(\{[V' \rightarrow [\cdot \text{nint}]\ V']\}) = I_{10}$